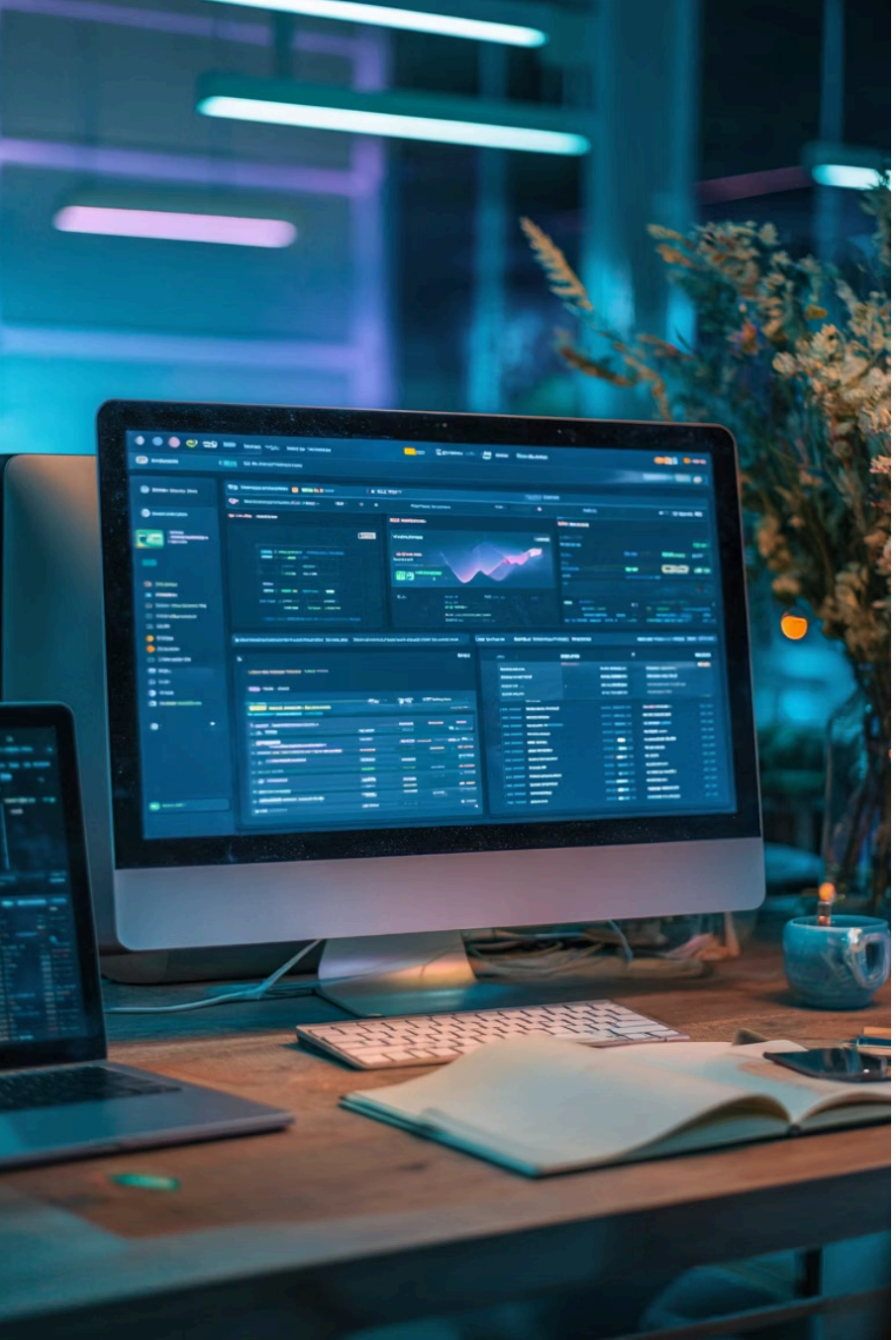

Customer Behaviour Analysis Report

SQL · Python · Power BI

PREPARED BY: GAURAV SARKAR

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Executive Summary

3,900

Customer Records

Across multiple US states

$59.76

Avg Purchase

Per transaction

3.75

Avg Review Rating

Out of 5.0

66%

Male Revenue Share

Top age groups: Adult & Senior

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Project Overview

Objectives

- Demographics & purchase behaviour
- Top products by revenue & ratings
- Discount & subscription impact
- Customer segmentation
- Interactive Power BI dashboard

Tools

- **PostgreSQL** — CTEs, Window Functions
- **Python** — Pandas, Matplotlib, Seaborn
- **Power BI** — DAX, Slicers, Maps

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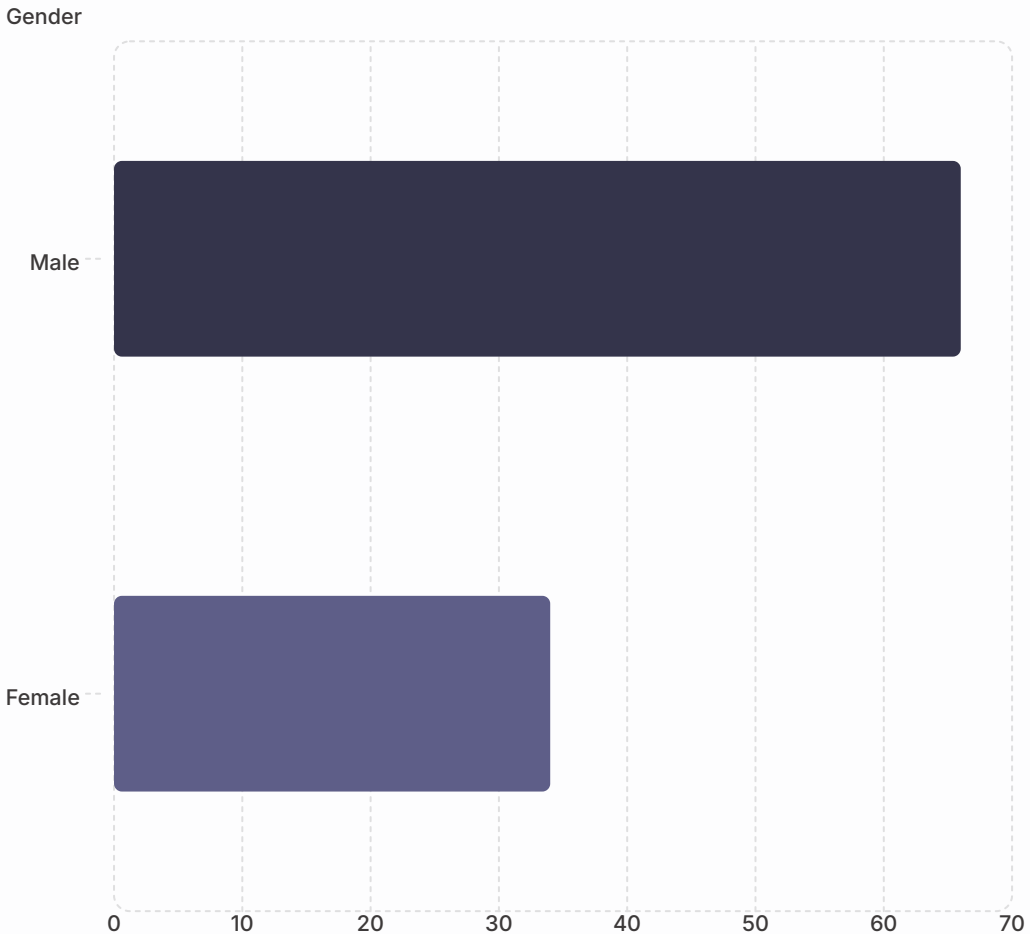
SQL Findings: Revenue & Discounts



Revenue by Gender

Male customers drive ~66% of total revenue. Female customers show competitive avg purchase amounts — key segment for targeted marketing.

High-Value Discount Users

Customers using discounts still **exceeded avg spend** — prime candidates for premium loyalty programs.

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SQL Findings: Subscriptions & Loyalty

Subscription Uplift

Subscribers spend more on average and generate higher total revenue

Repeat Buyers Subscribe

Customers with 5+ purchases show higher subscription propensity

Loyalty Segments

Majority are Returning or Loyal — focus on New → Loyal conversion

Healthy repeat purchase behaviour across the customer base.

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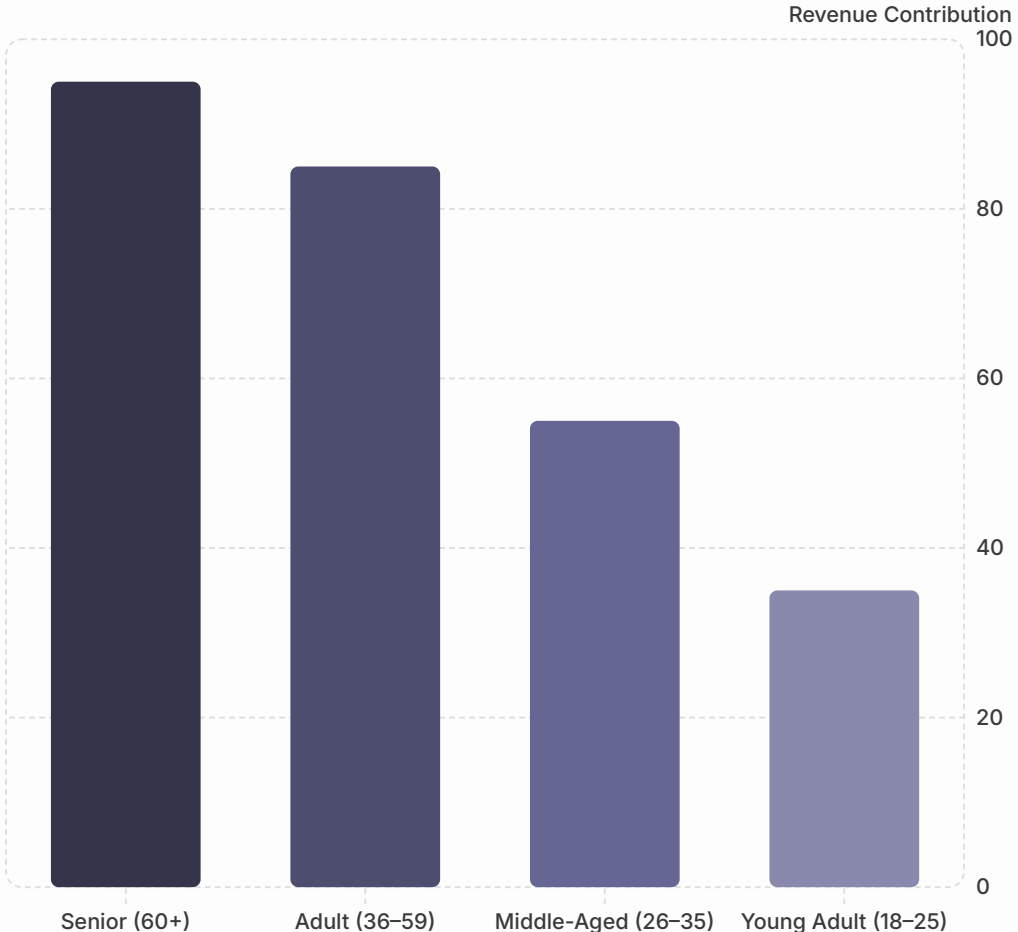
SQL Findings: Age & Products



Revenue by Age Group

Seniors & Adults drive most revenue. Young Adults represent a **growth opportunity** via digital engagement.

Top Products

Window functions ranked top 3 items per category — hero products for priority shelf space and ad spend.

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Python EDA Findings

Purchase Distribution

Most transactions $20–$100, uniform spread, no extreme outliers

Seasonal Patterns

Even across seasons; Winter slightly higher avg spend, colour preferences shift

Review Ratings

Avg 3.75/5 — products below 3.0 flagged for quality review

Correlation

Weak-moderate link between previous_purchases and subscription likelihood

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Power BI Dashboard

4-Page Architecture

Overview

KPIs, avg spend by age, revenue by season

Products

Top items, ratings, color heatmap, size analysis

Segments

Subscriber comparison, discount impact, gender split

Logistics

Shipping by age, payment donut, US state map

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Business Recommendations

→ Subscription Campaign

Target repeat buyers (5+ purchases) for conversion

→ Loyalty Discounts

Replace blanket promos with tiered loyalty offers

→ Premium for 36+

Focus premium product lines on Adult & Senior segments

→ Quality Audit

Review low-rated, high-volume SKUs

→ Express Upsell

Offer Express shipping at checkout to boost AOV

→ Geo Promotions

Target underserved US states with local campaigns

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Key Takeaways

Subscribers = Higher Value

Focus retention investment on subscribed customers

Repeat Buyers Convert

Natural candidates for subscription push

Age Segments Drive Revenue

Align product & marketing to demographic tiers

Full-stack workflow: SQL → Python EDA → Power BI dashboard with interactive slicers for self-service analytics.

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